

Importance Weighted Autoencoders on Dynamically Binarized MNIST

Reconstruction, DReG, and Progressive-Particle Training

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Deep Generative Models of Texts and Images

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Abstract

We reconstruct the one-stochastic-layer, $K = 50$ MNIST IWAE experiment of Burda et al. [2], then evaluate two meaningful extensions: DReG and progressive-particle DReG. The reconstruction reaches 87.804 test nats after 121 passes. DReG modestly improves likelihood to 87.480 nats but not KID. Progressive-particle DReG reaches 86.733 nats, KID 178.08 ± 21.76 , and trains in 14.0 minutes rather than about 71.5 minutes for fixed- K DReG. All numerical comparisons use seed 236; they are evidence for a practical profile, not a multi-seed superiority claim.

Part I

Paper Reconstruction

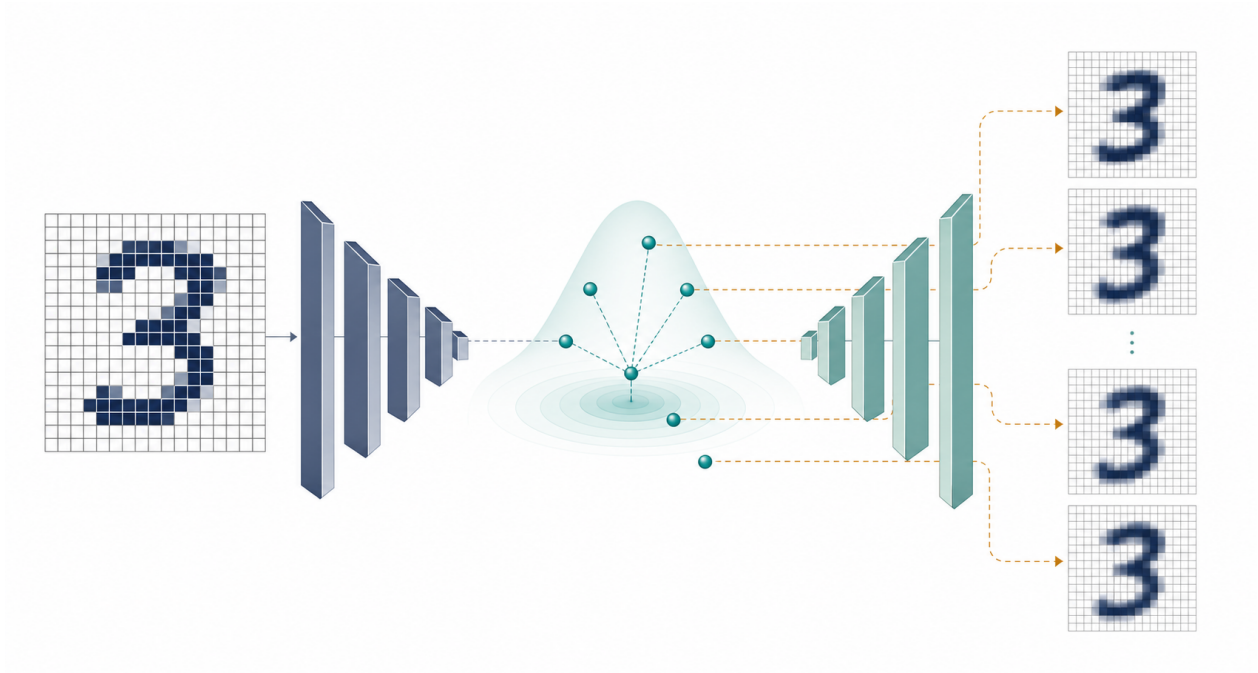


Figure 1: Conceptual visual abstract of the reconstruction pipeline. This generated illustration is explanatory only; all reported measurements are shown in the source-backed figures and tables below.

I.1 Original architecture

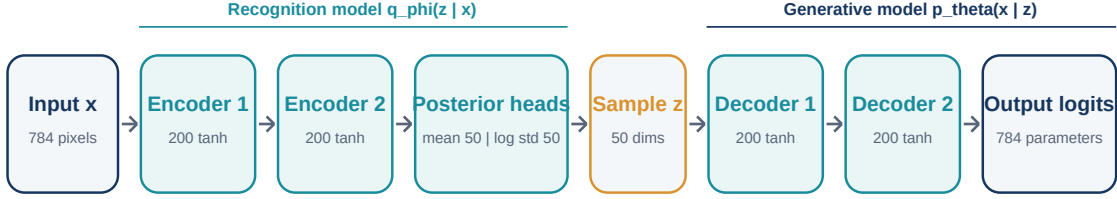
For a binarized image x , the model samples $z_{1:K} \sim q_\phi(z | x)$ and maximizes

$$\mathcal{L}_K(x) = \mathbb{E} \left[\log \frac{1}{K} \sum_{k=1}^K \frac{p_\theta(x, z_k)}{q_\phi(z_k | x)} \right]. \quad (1)$$

We minimize $-\mathcal{L}_{50}$ during training. Figure 2 and Table 1 give every architecture and training choice needed to reproduce the selected paper condition.

One-stochastic-layer IWAE: exact layer-by-layer architecture

The same architecture is used for the reconstruction, DReG, and progressive DReG.



Two 200-unit tanh layers in each network; a 50-dimensional diagonal Gaussian posterior; a factorized-Bernoulli decoder.

Figure 2: Source-backed architecture diagram. Both encoder and decoder use two 200-unit tanh hidden layers; the posterior is a 50-dimensional diagonal Gaussian and the decoder is Bernoulli.

I.2 Paper results and training set-up

The paper reports NLL in nats and active units. NLL is estimated here by $-\mathcal{L}_{5000}$ (lower is better); an active unit has posterior-mean variance above 0.01 across examples. The public MNIST dataset is downloaded by the project data module; the code and exact commands are in the repository README.

Table 1: Faithful reconstruction set-up and reported target.

Item	Paper / source code	Reconstruction
Data split	MNIST; source code reserves 400 examples	59,600 / 400 / 10,000
Training observations	dynamic Bernoulli binarization	matched
Architecture, latent size, particles	MLP, $z \in \mathbb{R}^{50}$, $K = 50$	matched
Initialization	Glorot; decoder log-odds output bias	matched
Optimizer	Adam, 10^{-3} , $\epsilon = 10^{-4}$	matched
Schedule	eight staged milestones through pass 3,280	matched milestones; run truncated at 121
Reported test NLL / active units	84.78 / 25	comparison target

I.3 Reconstruction results and loss graph

The loss graph uses the shared validation $-\mathcal{L}_{50}$ metric. The reconstruction is 3.024 nats above the paper’s NLL after only 3.69% of the original pass budget, while reproducing the reported 25 active units.

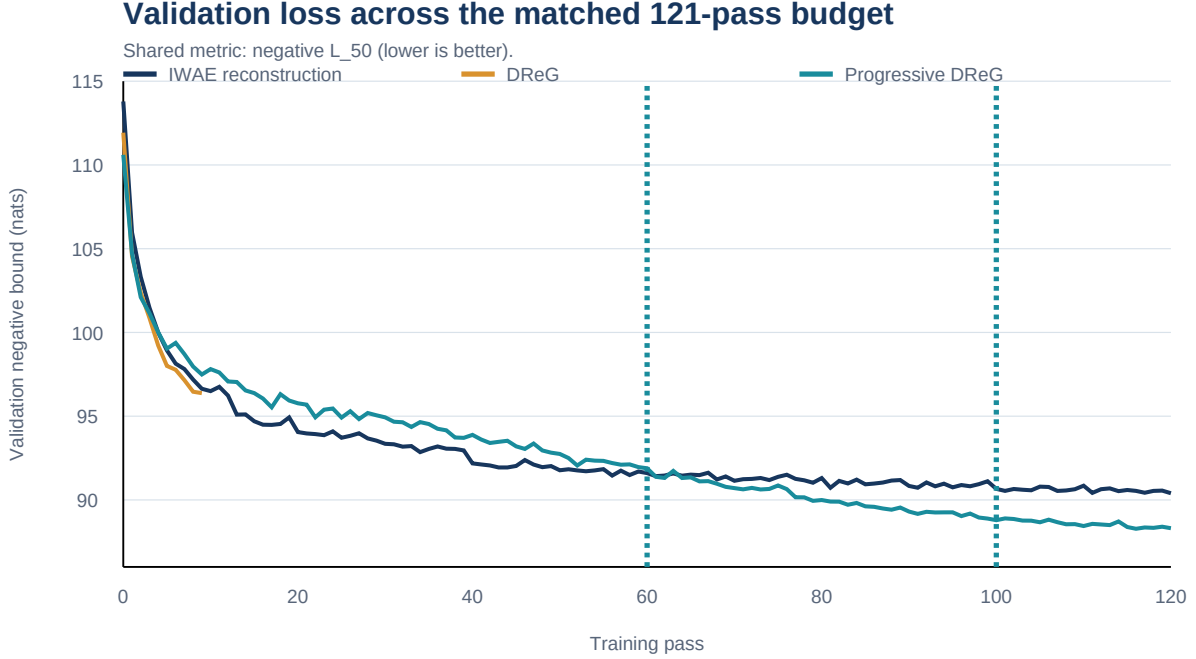


Figure 3: Validation loss graph for all completed 121-pass conditions. The two vertical guides mark the progressive model’s particle transitions.

Table 2: Paper result and reconstruction. NLL is the paper metric; lower is better.

Metric	Paper	Reconstruction (IWAE)
Training passes	3,280	121
Test $-\mathcal{L}_{5000}$ (nats)	84.78	87.804
Active units	25	25
KID	not reported	196.70 ± 23.05



Figure 4: Prior predictive decoder probabilities from the reconstruction. Brightness is Bernoulli mean; this is a qualitative check, not the primary metric.

Part II

DReG Improvement

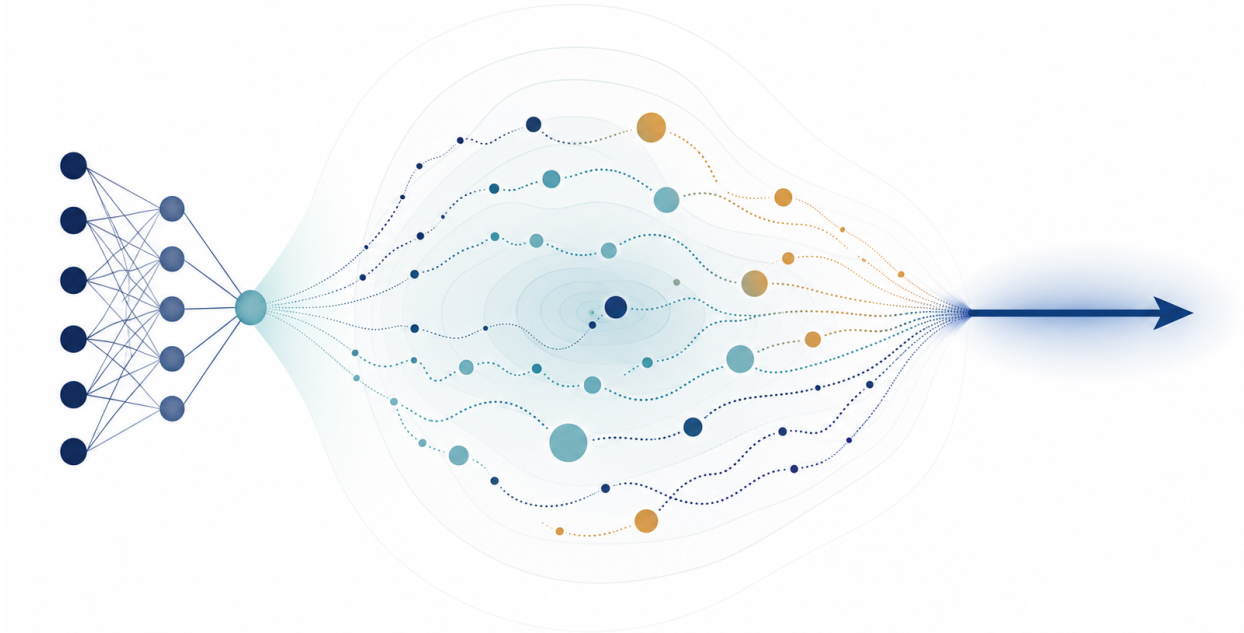


Figure 5: Conceptual visual abstract of the DReG extension. The generated illustration represents weighted latent paths and is not a measured gradient plot.

II.4 Improved architecture and training set-up

DReG is the assignment’s meaningful loss/gradient modification: it retains the generative model and IWAE objective but replaces the encoder’s high- variance gradient estimator with the doubly reparameterized estimator of Tucker et al. [3]. The encoder surrogate is

$$-\sum_k \text{sg}(\tilde{w}_k)^2 \log w_k, \quad \tilde{w}_k = \frac{w_k}{\sum_j w_j}, \quad (2)$$

where sg stops gradients through the normalized weight. The decoder keeps the ordinary IWAE gradient. Data, architecture, seed, initialization, $K = 50$, batch size 20, Adam settings, 121-pass budget, and the validation/test procedures remain fixed, so the changed estimator is the intended experimental variable.

II.5 Improved result: fixed- K DReG

The shared loss graph in Figure 3 shows the fixed- K DReG curve alongside the reconstruction and accelerated profiles.

Table 3: Matched seed-236 DReG comparison. The KID difference is not a run-to-run uncertainty estimate.

Metric	Reconstruction	Fixed- K DReG
Selected checkpoint pass	121	115
Best validation $-\mathcal{L}_{50}$	90.400	89.661
Test $-\mathcal{L}_{5000}$	87.804	87.480
Active units	25	26
KID	196.70 ± 23.05	206.80 ± 26.91
Training time (min)	74.2	approximately 71.5

DReG improves NLL by 0.323 nats but increases KID by 10.10 points. Thus it is a limited likelihood improvement, not a broadly superior model under this single-seed evidence.

Part III

Progressive-Particle Improvement

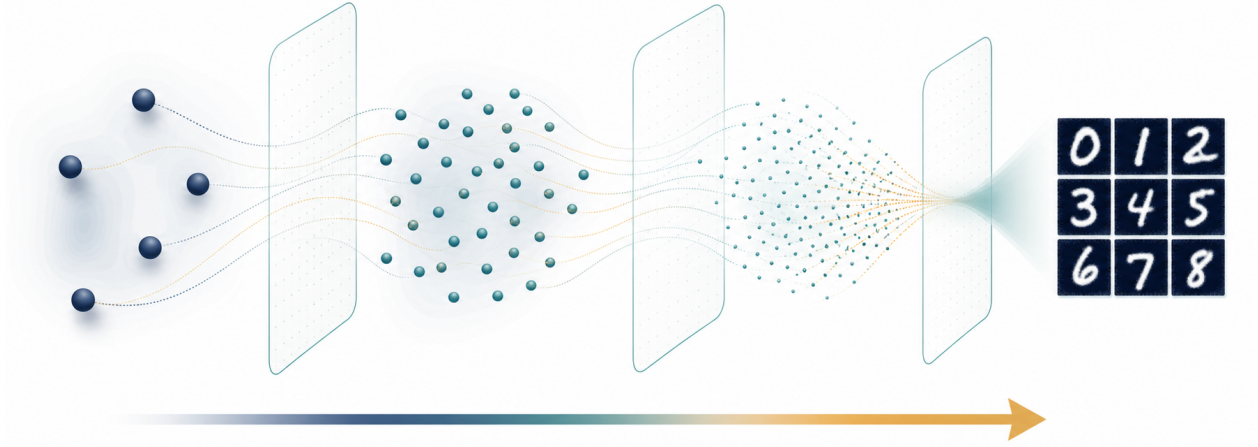


Figure 6: Conceptual visual abstract of progressive-particle training. The generated illustration conveys the curriculum; the exact schedule is in Figure 7.

III.6 Improved training set-up

This second improvement keeps DReG but changes the optimisation profile: batch size $20 \rightarrow 100$, particles $5 \rightarrow 10 \rightarrow 50$, and cosine learning-rate decay from 3×10^{-3} to 10^{-4} . Validation is always $K = 50$, making checkpoint selection directly comparable with fixed- K DReG. It is an optimisation extension, not a claim of paper-faithful training.

Progressive DReG training profile

The accelerated extension retains DReG and changes only the optimisation profile.

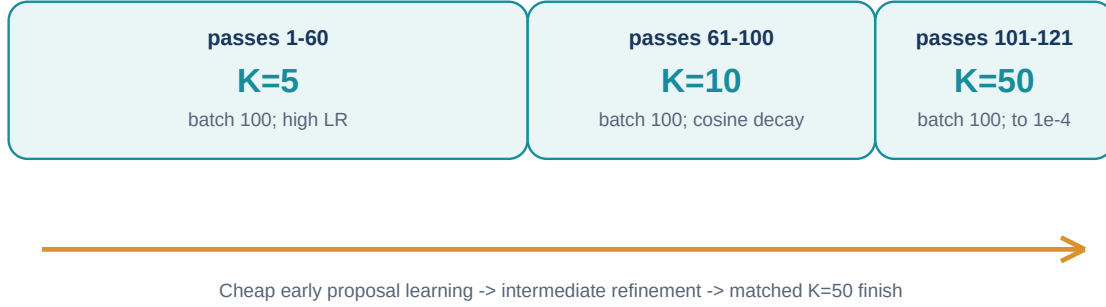


Figure 7: Exact accelerated training set-up. The final 21 passes use the same $K = 50$ objective as fixed- K DReG.

III.7 Improved results

KID is the required course metric in addition to the paper’s NLL. It is computed as Kernel Inception Distance [1] in a frozen 128-dimensional MNIST-classifier feature space. The shared classifier has 98.9% test accuracy; each result uses 10,000 real and generated images, 50 subsets of 1,000, and seed 236. Lower KID means closer feature distributions. The reported $\pm$ value is subset standard deviation, not replication uncertainty; the original paper did not report KID.

Table 4: Required comparison of paper, reconstruction, and improved models. Lower NLL, KID, and time are better.

Metric	Paper	Reconstruction	Fixed- K DReG	Progressive DReG
Training passes	3,280	121	121	121
Test $-\mathcal{L}_{5000}$	84.78	87.804	87.480	86.733
Active units	25	25	26	30
KID	n/a	196.70 $\pm$ 23.05	206.80 $\pm$ 26.91	178.08$\pm$21.76
End-to-end time (min)	n/a	74.2	approximately 71.5	14.0

Compact comparison of completed seed-236 runs

Lower is better for NLL, KID, and elapsed training time. KID bars show subset standard deviation.

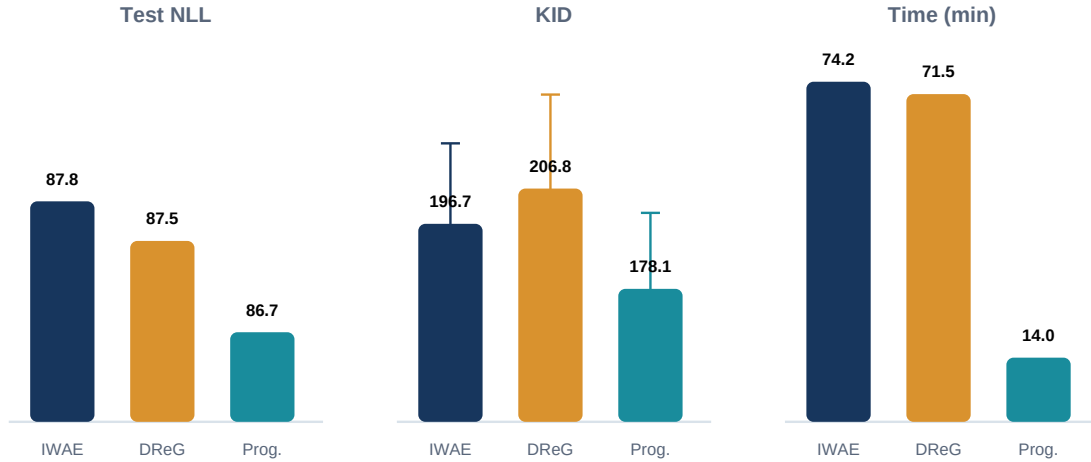


Figure 8: Generated comparison chart from the completed artifacts. Error bars appear only for KID subset standard deviation.

III.8 Discussion and lessons

- **What worked.** The original architecture and training core reproduce the paper’s 25 active units. DReG improves NLL, while the progressive profile improves NLL, KID, and elapsed time simultaneously.
- **What did not.** DReG alone worsened KID in this run. A truncated 121-pass reconstruction does not match the paper’s 3,280-pass NLL.
- **What we learned.** On this small IWAE, early $K = 50$ particles are expensive before the proposal is useful. Allocating particles over training is more consequential than changing the estimator alone.
- **Limitations.** All results use one seed. The fixed-DReG duration comes from interrupted-run artifacts, whereas progressive DReG is one uninterrupted run. Seeds 237 and 238, identical $\mathcal{L}_{5000}$ evaluation, KID, and a preregistered non-inferiority tolerance are needed before claiming reliable superiority.

Artifact availability and course-submission audit

The repository contains code for the reconstruction (`iwae train iwae`), the DReG and progressive improvements (`iwae train dreg` and `iwae train-fast`), $\mathcal{L}_{5000}$ evaluation, sample generation, the reusable MNIST feature classifier, and KID evaluation. The data module downloads public MNIST automatically from <http://yann.lecun.com/exdb/mnist/>. The relevant saved artifacts are:

- accelerated checkpoint: `outputs/fast-dreg/checkpoints/epoch-0116.ckpt`;
- shared KID classifier: `outputs/classifier/checkpoints/epoch-04.ckpt`.

These cover the reconstruction code, improved-model code, public dataset access, and PDF report required by the Part 2 brief.

References

- [1] Mikołaj Bińkowski, Danica J. Sutherland, Michael Arbel, and Arthur Gretton. Demystifying MMD GANs. In *International Conference on Learning Representations*, 2018. URL <https://arxiv.org/abs/1801.01401>.
- [2] Yuri Burda, Roger Grosse, and Ruslan Salakhutdinov. Importance weighted autoencoders. In *International Conference on Learning Representations*, 2016. URL <https://arxiv.org/abs/1509.00519>.
- [3] George Tucker, Dieterich Lawson, Shixiang Gu, and Chris J. Maddison. Doubly reparameterized gradient estimators for monte carlo objectives. In *International Conference on Learning Representations*, 2019. URL <https://arxiv.org/abs/1810.04152>.